6CCSACCM: Cooperation, Coordination and Multi-Agent Systems

Week 7

Fair division (part I)

Maria Polukarov

Fair division

- Fair division is the problem of dividing one or several goods amongst two or more agents in a way that satisfies a suitable fairness criterion.
- That is, fair division may be considered part of the larger research area of multiagent resource allocation.
- What is special about fair division is the explicit focus on fairness concerns.

Types of goods

- Divisible or indivisible
- Static or perishable/consumable
- Sharable or not
- Single or multiple units

Cake cutting: mathematical model

We model the cake as the unit interval $[0, 1]$ and define the following:

- A set of players $N = \{1, \ldots, n\}$.
- Pieces of cake $X_1, \ldots, X_m \subseteq [0, 1]$, each of which is a finite union of disjoint subintervals. The pieces are themselves disjoint, and their union gives the whole cake: $\forall i, j, X_i \cap X_j = \emptyset$; $\bigcup_{i=1}^{m} X_i = [0, 1]$.

Cake cutting: mathematical model

Each player $i \in N$ has a non-negative, absolutely continuous valuation function $v_i(X)$ over pieces of cake, which satisfies the following properties:

- **Additivity:** If X and Y are disjoint, then $v_i(X \cup Y) = v_i(X) + v_i(Y)$.
- **Normalization:** $v_i([0, 1]) = 1$.
- **Divisibility:** Given an interval I and $\lambda \in [0, 1]$, there exists a subinterval $I' \subseteq I$ such that $v_i(I') = \lambda v_i(I)$.

Cake cutting: mathematical model

Goal: an allocation $(A_1, \ldots, A_n)$

- A_i is the piece of cake allocated to agent i
- pieces should be disjoint: $A_i \cap A_j = \emptyset$ for all $i \neq j$
- an allocation is complete if $\bigcup_{i=1}^{n} A_i = [0, 1]$

Cake cutting: example

- $v_1([p,q]) = 4|q-p|$ for each $0 \leq p < q \leq 1/4$, $v_1([1/4,1]) = 0$
- $v_2([p,q]) = |q-p|$ for each $0 \leq p < q \leq 1$

- $A_1 = [0, 1/2]$, $A_2 = [1/2, 1]$
- $A_1 = [0, 1/4]$, $A_2 = [1/4, 1]$
- $A_1 = [0, 1/5]$, $A_2 = [1/5, 1]$

Solution concept: proportionality

An allocation is **proportional** if each agent values their share as at least $1/n$.

- $A_1 = [0, 1/2]$, $A_2 = [1/2, 1]$ ✓
- $A_1 = [0, 1/4]$, $A_2 = [1/4, 1]$ ✓
- $A_1 = [0, 1/5]$, $A_2 = [1/5, 1]$ ✓

Solution concept: proportionality

An allocation is **proportional** if each agent values their share as at least $1/n$.

- $A_1 = [0, 1/2]$, $A_2 = [1/2, 1]$ ✓
- $A_1 = [0, 1/4]$, $A_2 = [1/4, 1]$ ✓
- $A_1 = [0, 2/3]$, $A_2 = [2/3, 1]$ ✗

Solution concept: envy-freeness

An allocation is **envy-free** if no agent envies another one: $v_i(A_i) \geq v_i(A_j)$ for all i, j.

- $A_1 = [0, 1/2]$, $A_2 = [1/2, 1]$ ✓
- $A_2 = [0, 1/2]$, $A_1 = [1/2, 1]$ ✗

Solution concept: equitability

An allocation is **equitable** if all agents have the same value: $v_i(A_i) \geq v_j(A_j)$ for all i, j.

- $A_1 = [0, 1/2]$, $A_2 = [1/2, 1]$ ✗
- $A_1 = [0, 1/4]$, $A_2 = [1/4, 1]$ ✗
- $A_1 = [0, 1/5]$, $A_2 = [1/5, 1]$ ✓

Solution concept: Pareto-optimality

An allocation $(A_1, \ldots, A_n)$ is Pareto-optimal if there is no other allocation $(B_1, \ldots, B_n)$ such that $v_i(B_i) \geq v_i(A_i)$ for each i and $v_i(B_i) > v_i(A_i)$ for some i.

- $A_1 = [0, 1/2]$, $A_2 = [1/2, 1]$ ✗
- $A_1 = [0, 1/4]$, $A_2 = [1/4, 1]$ ✓

Relationship between solutions

Theorem

For two players, an allocation is proportional if and only if it is envy-free.

Proof.

- *Envy-freeness implies proportionality:* Without loss of generality, if player 1 does not envy player 2, then $v_1(A_1) \geq v_1(A_2)$. Since $v_1(A_1) + v_1(A_2) = 1$, this implies $v_1(A_1) \geq 1/2$.
- *Proportionality implies envy-freeness:* If the allocation is proportional, then $v_1(A_1) \geq 1/2$. Since $v_1(A_1) + v_1(A_2) = 1$, this implies $v_1(A_1) \geq v_1(A_2)$, so player 1 does not envy player 2.

$\square$

Relationship between solutions

Theorem

For three or more players, envy-freeness implies proportionality, but the converse is not true.

Proof.

- *Envy-freeness implies proportionality:* Let A_i be what player i values most among $A_1, \ldots, A_n$. Then, since $\sum_{j=1}^{n} v_i(A_j) = 1$, it must be that $v_i(A_i) \geq 1/n$. Since the allocation is envy-free, player i must receive A_i, so the allocation is proportional.
- *Proportionality does not imply envy-freeness:* Consider the case of three players. Let player 1's allocation A_1 be such that $v_1(A_1) = 1/3$, but $v_1(A_2) = 1/3 + \varepsilon$ and $v_1(A_3) = 1/3 - \varepsilon$ for some small $\varepsilon > 0$. Then the allocation is proportional, but player 1 envies player 2.

$\square$

Computation

- Do fair solutions always exist?
- Can we find them efficiently?
  - What are some algorithms that can achieve proportional or envy-free allocations?
  - How do we measure the complexity of these algorithms?
- **Proportionality:** yes, with a simple algorithm!
- **Envy-freeness:** yes, throw out the cake!!
- **Envy-freeness and completeness:** yes, but the runtime of the best algorithm for n agents is $O(n^{n^{n^{n^{n}}}})$!!!

Two agents: Cut-and-Choose

1. Player 1 cuts the cake into two pieces X_1 and X_2 they value equally: $v_1(X_1) = v_1(X_2) = 1/2$.
2. Player 2 chooses the piece they prefer, and player 1 receives the other piece.

We claim that this is envy-free, and hence also proportional.

Proof.

Player 1 cuts the cake into two pieces they value equally, so they will not envy player 2. Player 2 then chooses the piece they value most, so they will not envy player 1. Thus, the allocation is envy-free. $\square$

Two agents: Cut-and-Choose

The Cut-and-Choose allocation is not necessarily equitable.

- agent 1 likes chocolate, agent 2 likes everything
- if agent 1 cuts first, $A_1 = [0, 1/4], A_2 = [1/4, 1]$
- if agent 2 cuts first, $A_1 = [0, 1/2], A_2 = [1/2, 1]$

Measuring complexity: The Robertson-Webb query model

To implement Cut-and-Choose, agents should be able to cut a piece of a given value and evaluate a piece.
We measure the complexity of cake-cutting algorithms using the Robertson-Webb model, which defines two types of operations:

- **Evaluation Query:** $Eval_i(x, y)$ returns $v_i([x, y])$. It asks "how much does player i value the piece $[x, y]$?"
- **Cut Query:** $Cut_i(x, \alpha)$ returns y such that $v_i([x, y]) = \alpha$. It asks "Please cut a piece starting at x that player i values at α."

Under the Robertson-Webb model, the Cut-and-Choose algorithm requires two operations.

Dubins-Spanier algorithm: Proportionality for n players

A ‘moving-knife’ algorithm that ensures proportionality:

1. A referee moves a knife continuously across the cake.
2. When any player finds the segment left of the knife worth at least $1/n$, they shout “stop”. They receive that segment and are removed from the game. Ties are broken arbitrarily.
3. The process repeats with the remaining players until one player is left, who receives the remainder

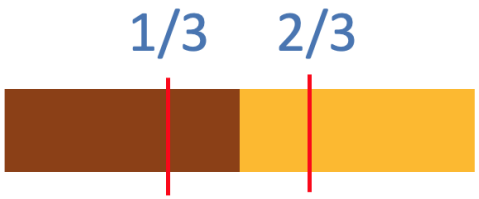
Dubins-Spanier algorithm: Proportionality for n players

The Dubins-Spanier allocation is proportional.

Proof.
Any player who shouted “stop” received a segment worth at least $1/n$ to them, so they received their proportional share. The last player did not shout “stop” at all, or shouted “stop” at least once at the same time as some other player, so they must have valued the previous $n-1$ pieces each at at most $1/n$, and hence the remaining piece they receive is valued at least $1/n$. $\square$

Dubins-Spanier algorithm: envy-freeness?

For $n > 2$, Dubins-Spanier protocol may fail to produce an envy-free allocation.



- agent 1 likes everything, agents 2 and 3 like sponge only
- agent 1 shouts "Stop!" at $1/3$ and gets $[0, 1/3]$
- agents 2 and 3 shout "Stop!" at 2/3
  - one of them (say, agent 2) gets $[1/3, 2/3]$, the other gets $[2/3, 1]$
- agent 2 envies agent 3: agent 3 gets more sponge

Also: not equitable, not Pareto-optimal.

Dubins-Spanier algorithm: complexity

Under the Robertson-Webb model, the Dubins-Spanier algorithm requires $\Theta(n^2)$ operations.

Proof.

The number of cuts is $n + (n-1) + \ldots + 2$, so the total complexity is $\Theta(n^2)$. $\square$

Even-Paz algorithm: optimal proportionality for n players

For ease of exposition, we describe the algorithm for $n = 2^k$ players:

1. Denote the remainder of our cake as $[s, t]$. At the first iteration, initialize $s = 0$ and $t = 1$.
2. If $n = 1$, the player receives $[s, t]$. Otherwise:
3. Each player i makes a mark z_i that divides $[s, t]$ into two equal-valued pieces: let $v_i = Eval_i(s, t)$, then $Cut_i(s, \frac{1}{2}v_i) = z_i$.
4. Let $z*$ be the $n/2$th mark from the left.
5. Recurse on the left half $[s, z*]$ with the $n/2$ players who marked $z_i \leq z*$, and the right half $(z*, t]$ with the remaining $n/2$ players.

Even-Paz algorithm: optimal proportionality for n players

The Even-Paz algorithm achieves proportionality in $O(n \log n)$ operations in the Robertson-Webb model.

Proof.
The recursion depth is $O(\log n)$ since we halve the number of players at each step, and each step requires $O(n)$ operations. Thus, the total complexity is $O(n \log n)$. $\square$

It is also known that:

Theorem
Any proportional cake-cutting algorithm requires $\Omega(n \log n)$ operations in the Robertson-Webb model.

Hence, the Even-Paz algorithm is optimal under the Robertson-Webb model.

Even-Paz algorithm: optimal proportionality for n players

The Even-Paz allocation is proportional.

Proof.

We can prove by induction that after k steps, the players sharing a piece of cake value it at least at $1/2^k$. For $k = 0$, this is true because the piece is initially the whole cake. For the induction step, if the piece of cake at step k is $[x, y]$, at step $k + 1$, players are sharing a piece valued at least $v_i([x, y])/2$. Therefore, if a player's value was at least $1/2^k$, now it is at least $1/2^{k+1}$. The process ends after $\log n$ steps with a single player being allocated a piece. We conclude that the value of each player is at least $1/2^{\log n} = 1/n$. $\square$

Selfridge-Conway algorithm: envy-freeness for three players

- In general, envy-free cake-cutting is algorithmically more challenging than proportional cake-cutting.
- This algorithm gets an envy-free allocation for three players.
- The algorithm takes place in three stages.

Selfridge-Conway algorithm: envy-freeness for three players

Stage 0:

1. Player 1 cuts the cake into three pieces they value equally.
2. Player 2 trims the largest piece such that the two largest pieces are now equal in their eyes. There are now four pieces: the two most valuable equal pieces (one of which was trimmed), the trimming, and the remaining original piece.
3. Let Cake 2 be the trimming, and Cake 1 be the other three pieces.

Selfridge-Conway algorithm: envy-freeness for three players

Stage 1: We now divide Cake 1 first.

1. Player 3 chooses their highest-valued piece in Cake 1.
2. If Player 3 did not choose the trimmed piece, Player 2 is allocated the trimmed piece. Otherwise, if Player 3 chose the trimmed piece, Player 2 chooses their highest-valued piece out of the two remaining ones.
3. The remaining piece is allocated to Player 1.

Selfridge-Conway algorithm: envy-freeness for three players

Stage 2: We finally divide Cake 2.
At this point, either Player 2 or 3 received the trimmed piece. Among players 2 and 3, let T be the player that received the trimmed piece, and T' be the other player.

1. Player T' divides Cake 2 into three pieces they value equally.
2. Players T, 1, then T' choose one of the three pieces each in that order.

Selfridge-Conway algorithm: envy-freeness for three players

We now prove that the Selfridge-Conway allocation is envy-free.
For each player, since valuations are additive, if we show that they are envy-free on Cake 1 and Cake 2 then they are envy-free overall. We will do this for each player.

- **Player T'.** This is clearly true on Cake 2 since they divided it equally. On Cake 1, if T' is Player 3, they got to choose their highest-valued piece out of all 3 original pieces, so they are envy-free. If T' is Player 2, recall that they trimmed the two largest pieces such that they were equal in Stage 0, and they received one of those largest pieces in Stage 1. Hence, they are envy-free, since they received their highest-valued piece overall.

Selfridge-Conway algorithm: envy-freeness for three players

- **Player T.** On Cake 2, they chose their highest-valued piece first, so they are envy-free. On Cake 1, the argument is the same as above: Player 3 and 2 are both envy-free, so Player T is certainly envy-free.
- **Player 1.** On Cake 1, since they did not receive the trimmed piece, then they received a piece certainly worth $1/3$ as they cut the cake equally in Stage 0. On Cake 2, Player 1 does not envy T' because they choose before T'. The only concern left is T. But notice that T received the trimmed piece from Cake 1, which can only be worth at most $1/3$ if T then receives the entire Cake 2! In this way, Player 1 has an irrevocable advantage over T in the allocation of Cake 2, so they do not envy T. $\square$

Complexity of Envy-Free Algorithms

It has been shown that an envy-free cake-cutting algorithm always exists.

Theorem (Brams and Taylor 1995)

There exists an envy-free cake-cutting algorithm for any number of players under the Robertson-Webb model.

However, by changing the valuation function of the players, the complexity of the algorithm described by Brams and Taylor can be made arbitrarily high, i.e. cannot be bounded as a function of n.

Complexity of Envy-Free Algorithms

It was later shown that a bound does exist, but it is extremely high.

Theorem (Aziz and Mackenzie 2016)

A bounded envy-free algorithm exists for any n, but its complexity is extremely high: $O(n^{nnnnn})$.

A lower bound of $\Omega(n^2)$ was also shown for envy-free cake-cutting algorithms.

Theorem (Procaccia 2009)

Any envy-free cake-cutting algorithm requires $\Omega(n^2)$ *operations in the Robertson-Webb model.*

THANK YOU!